

MECHANICAL DESIGN OF A SIMPLE ROBOTIC DOG

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Project Objective

The objective of this task is to create an initial mechanical design of a simple robotic dog. The design focuses on the body structure, moving joints, degrees of freedom, actuator selection, fastening method, mass properties, torque estimation, and expected mechanical issues.

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Body Shape and Structure

The robot was designed with a rectangular main body, four legs, a neck section, and a front robotic arm. The main body acts as the chassis and supports the leg assemblies. The long rectangular body helps distribute the load along the robot length, while the four legs provide balance and support during standing and walking.

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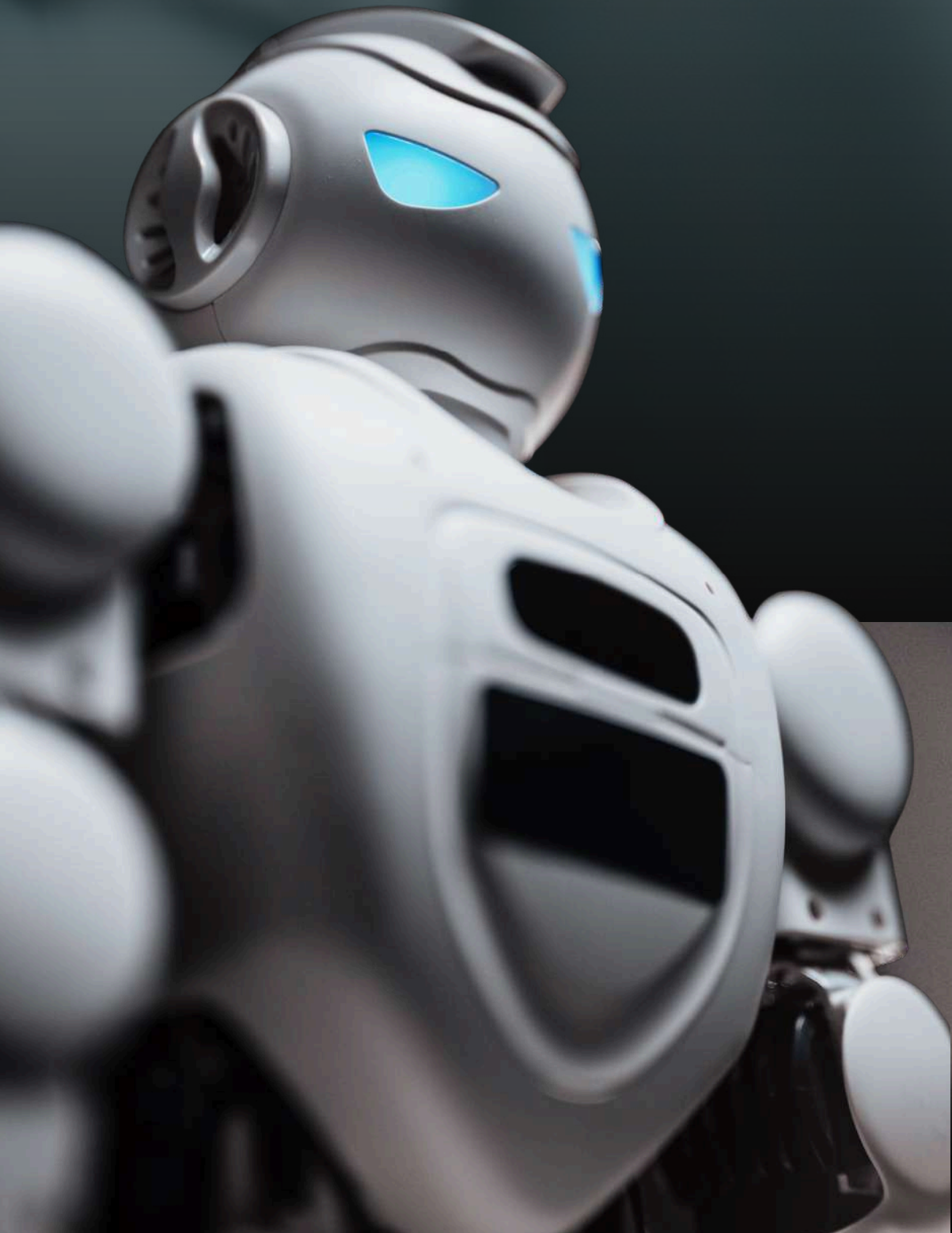
Main Components

Component	Function in the design
Main body	Central chassis that carries all mechanical parts.
Front and rear legs	Support the robot and provide the walking mechanism.
Neck section	Connects the front arm to the body.
Front arm	Simple manipulator mechanism attached to the front.
Revolute joints / pins	Allow rotation between moving links.

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Dimensions and Mass Properties

Property	Value
Overall length	133.57 mm
Overall width	76.49 mm
Overall height	109.31 mm
Model volume	60,477.44 mm ³ = 60.48 cm ³
Center of mass	X = -0.86 mm, Y = -7.97 mm, Z = 53.26 mm
Assumed material for calculation	PLA plastic
PLA density used	1.24 g/cm ³
Estimated mass	74.99 g = 0.0750 kg



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Torque Calculation for Shoulder Joint

The shoulder link length measured in Tinkercad is 20 mm. Torque is calculated using the worst-case static condition, where the load acts perpendicular to the shoulder link. The model volume was extracted from the provided CAD file, and the mass was estimated using PLA density

Volume = 60.48 cm³

Density of PLA = 1.24 g/cm³

Mass = Density × Volume = 1.24 × 60.48 = 74.99 g = 0.0750 kg

Force = m × g = 0.0750 × 9.81 = 0.736 N

Lever arm distance = 20 mm = 0.020 m

Torque = Force × Distance = 0.736 × 0.020 = 0.01471 N·m

Torque = 1.47 N·cm = 0.15 kg·cm

Item	Calculated value
Shoulder length / lever arm	20 mm = 0.020 m
Estimated robot mass using PLA	0.0750 kg
Weight force	0.736 N
Required static torque	0.01471 N·m
Required static torque	1.47 N·cm
Required static torque	0.15 kg·cm
With safety factor 2	0.30 kg·cm
With safety factor 3	0.45 kg·cm

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Motor Selection

Based on the calculated shoulder torque, the minimum static torque is approximately 0.15 kg·cm. After applying a safety factor of 3, the recommended torque becomes about 0.45 kg·cm. Therefore, a small servo motor rated at 1 kg·cm or higher is acceptable for a simple prototype. A common SG90 micro servo, which is typically rated around 1.8 kg·cm, is suitable for this educational model

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Joints and Degrees of Freedom

The moving connections are modeled as revolute joints. The shoulder joint provides rotational movement for the leg. The neck/front arm section can also rotate as a simple revolute mechanism. For the initial mechanical concept, the design can be described as a multi-joint robotic dog with rotational degrees of freedom at the leg and neck joints

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Fastening Method

The proposed fastening method is to use cylindrical pins or small screws through the revolute joints. This allows the moving parts to rotate while keeping the parts assembled. Fixed body parts can be attached using screws or adhesive depending on the final manufacturing method

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Expected Mechanical Problems

Expected issues include joint wear due to repeated motion, loosening of screws or pins, instability on uneven surfaces, and reduced performance if the payload is increased. High friction in the joints can also increase the torque required from the servo motors

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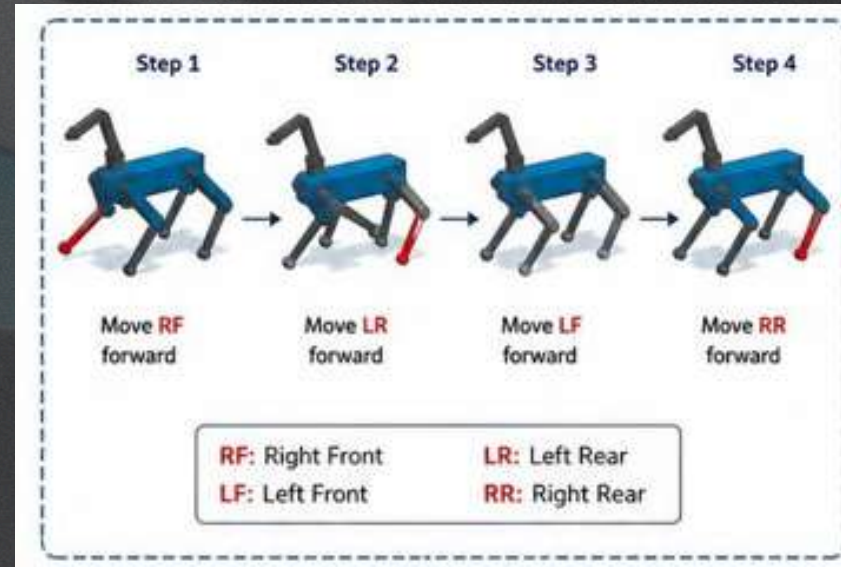
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Proposed walking Method

The robot is expected to use a crawl gait, where only one leg moves at a time while the other three legs remain in contact with the ground. This walking method provides high stability and is suitable for a simple four-legged robotic dog prototype. Crawl gait sequence (example - starting with right front leg):

- 1.Right Front (RF) leg moves forward.
- 2.Left Rear (LR) leg moves forward
- 3.Left Front (LF) leg moves forward.
- 4.Right Rear (RR) leg moves forward.
- 5.Repeat the sequence.



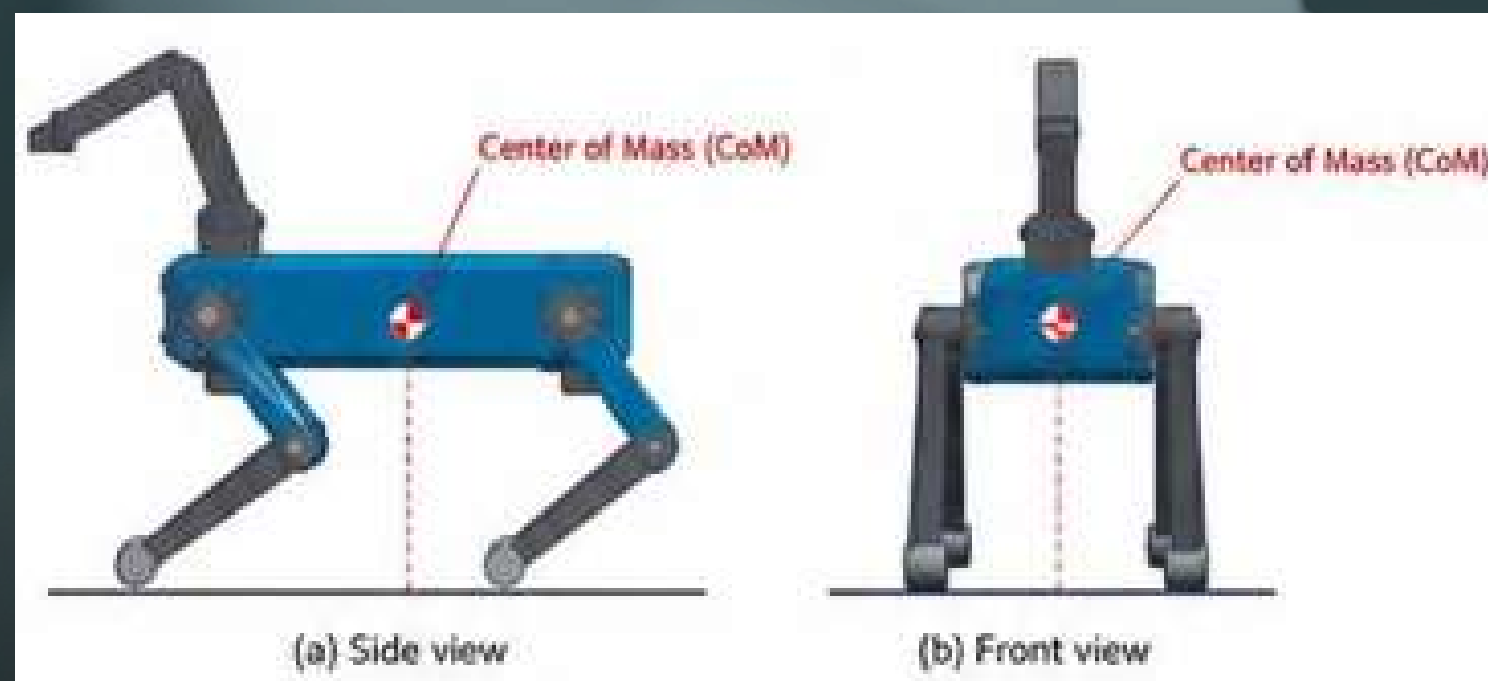
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Center of Mass Visualization

The center of mass (CoM) of the robot is shown in the figures. It is located near the middle of the main body to ensure good balance during walking.

Center of mass coordinates:

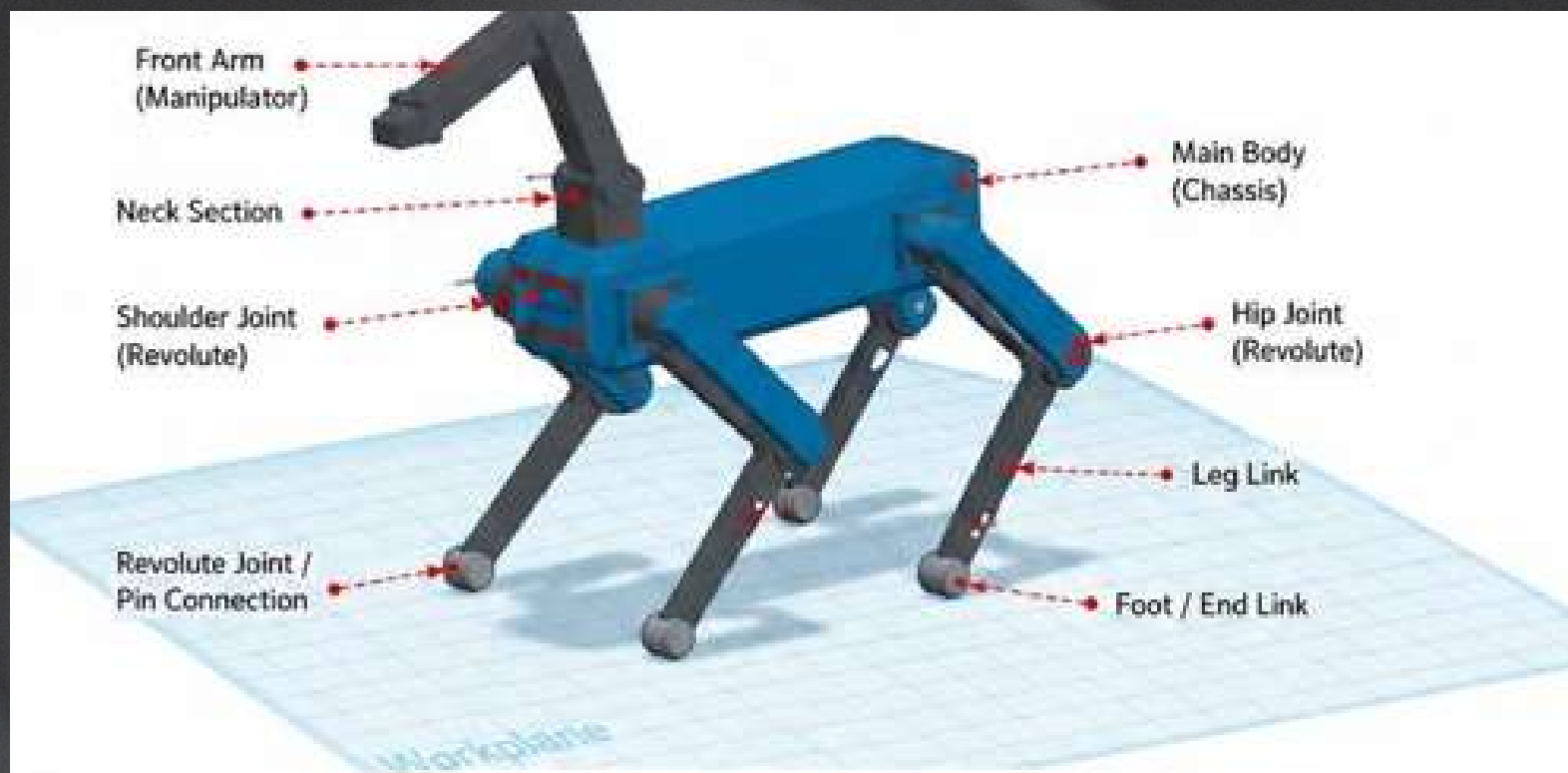
$X = -0.86 \text{ mm}$ / $Y = -7.97 \text{ mm}$ / $Z = 53.26 \text{ mm}$



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Labeled components



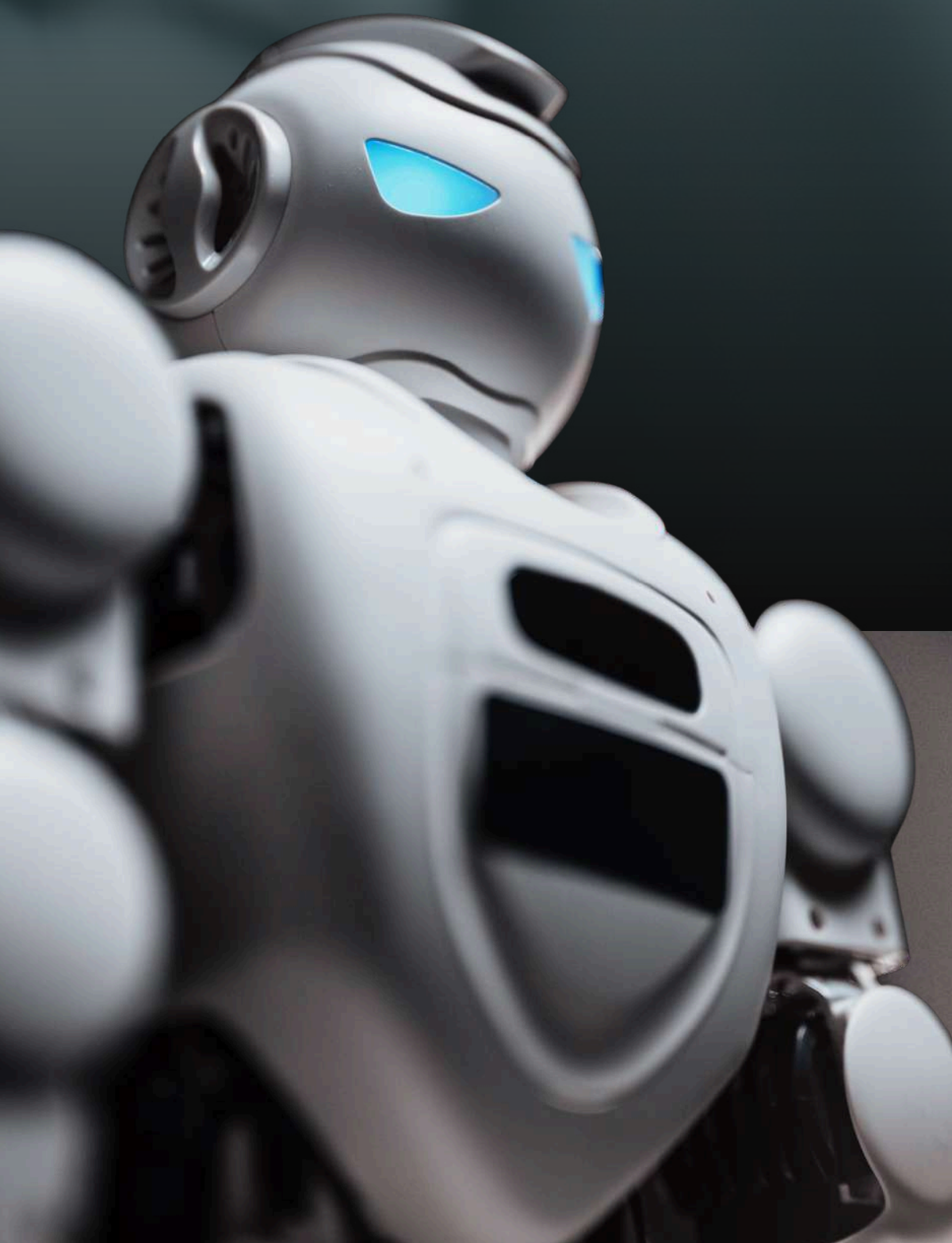
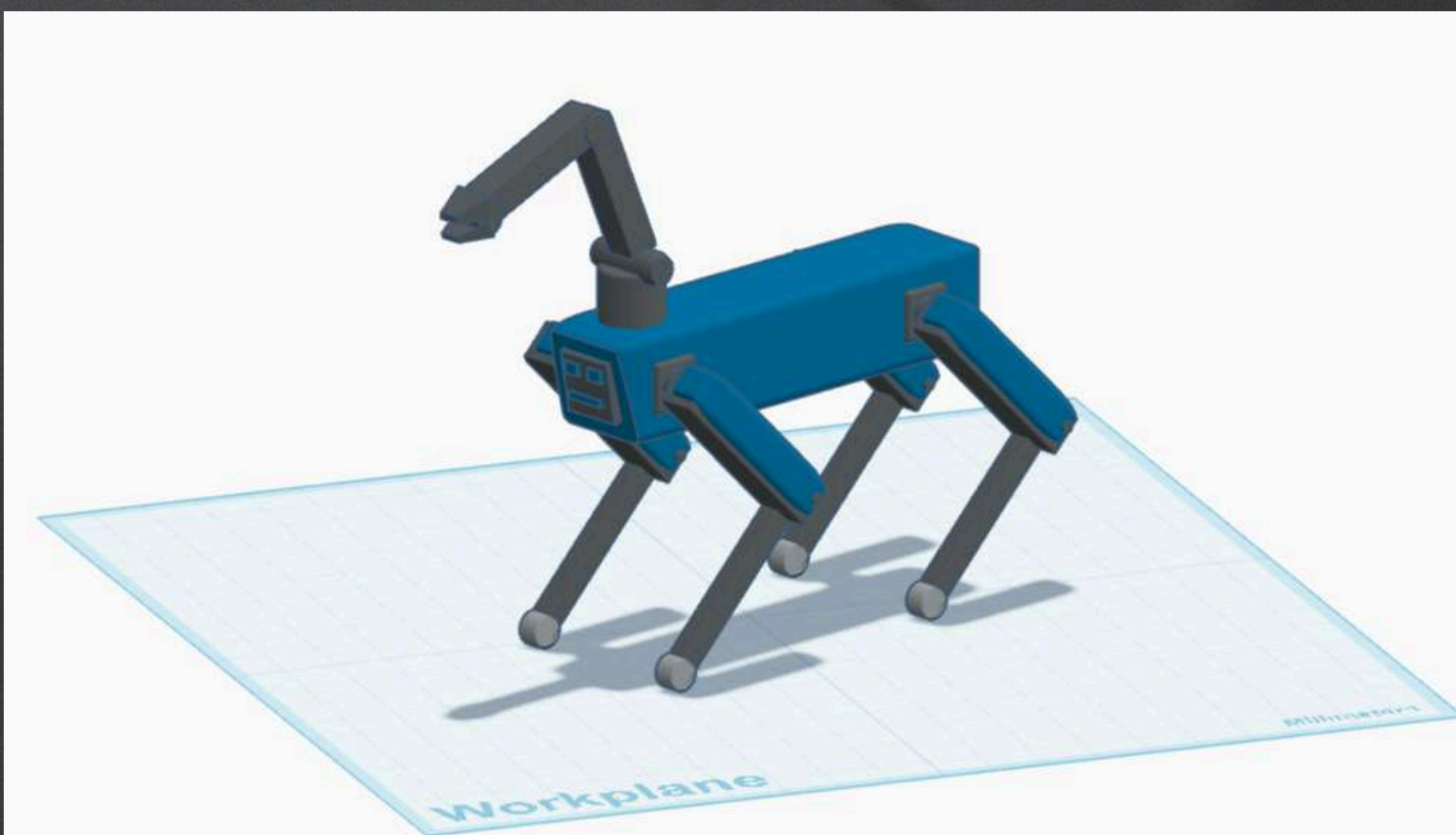
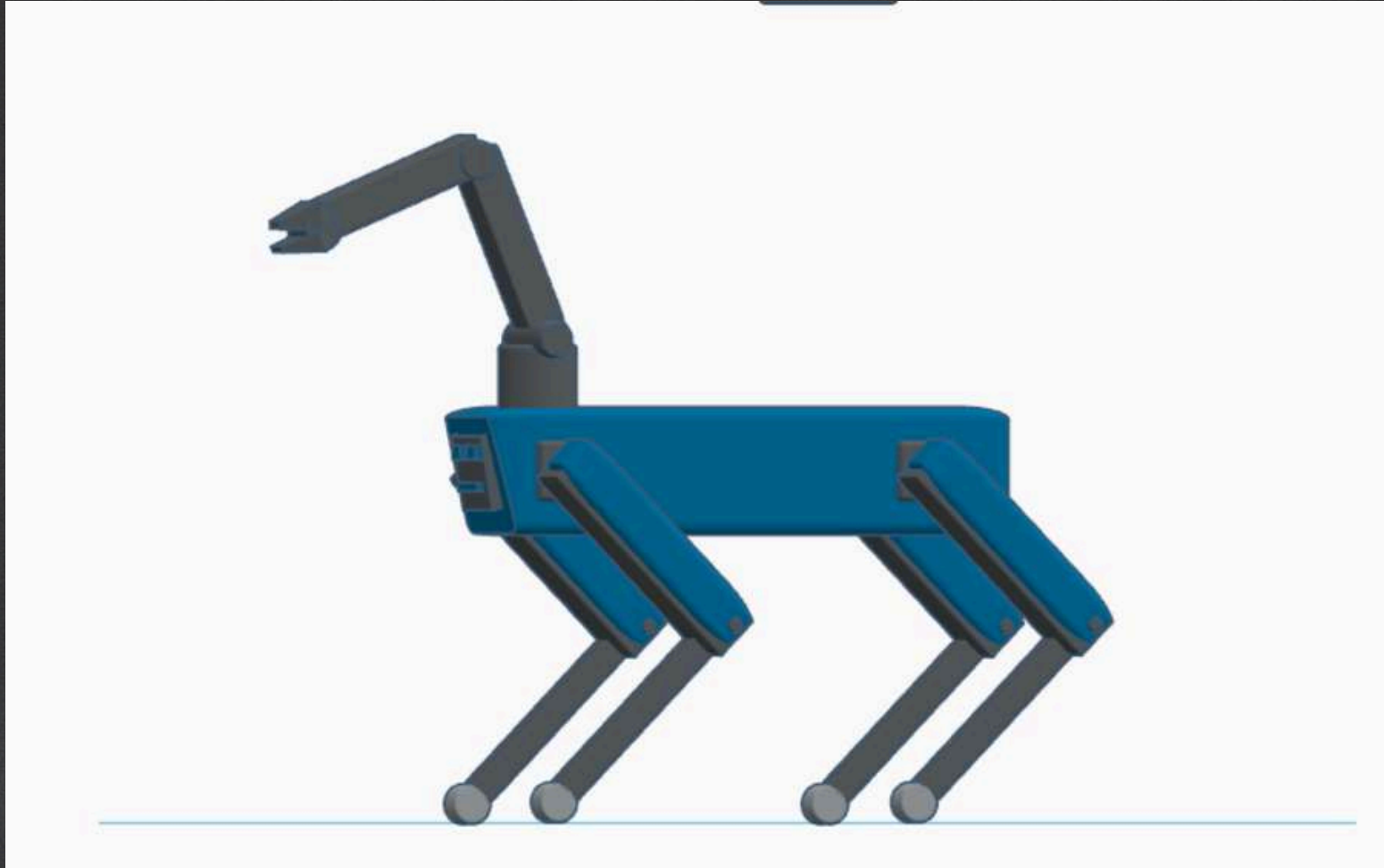
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Conclusion

The designed robotic dog satisfies the requirements of the first mechanical design task. The report includes the body structure, main parts, dimensions, center of mass, material assumption, mass estimation, torque calculation, motor selection, fastening method, and expected mechanical problems. The model is suitable as a first prototype for future development using servo motors and embedded control

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